Week 8: 19th to 23rd OCtober

Subject / الموضوع: Science **Unit Title / عنوان الوحدة: Heredity and Natural Selection – الوراثة والانتقاء الطبيعي** **Grade / الصف: 8**

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1 الحصة الأولى **Genes, Mutations and Traits** الجينات والطفرات والصفات	**Analyze:** I can analyze how a mutation in a gene changes a protein and explain why its effect on an organism's traits can be harmful, beneficial or neutral.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.1 Activate prior knowledge using organizers, links, and pre-teaching. 8.2 Vary challenge level with differentiated tools, resources, and difficulty.	Kindly check Toddle for the lesson PPT and practice sheet.
Lesson 2 الحصة الثانية **Natural Selection** الانتقاء الطبيعي	**Construct an explanation:** I can use evidence to explain how variations in traits increase some individuals' chances of surviving and reproducing in a specific environment.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 3.2 Highlight big ideas, patterns, and critical features using organizers and examples. 8.3 Foster collaboration and community with structured group roles and supports.	Kindly check Toddle for the lesson PPT and practice sheet. PhET simulation: phet.colorado.edu/en/simulations/natural-selection
Lesson 3 الحصة الثالثة **Artificial Selection** الانتقاء الاصطناعي	**Compare:** I can compare natural and artificial selection and explain how humans choose desirable traits to change organisms over many generations.	7.2 Make learning relevant, authentic, and connected to students' backgrounds. 3.3 Guide processing with explicit steps, chunking, and interactive models. 6.3 Facilitate organization with graphic organizers, prompts, and note-taking guides.	Kindly check Toddle for the lesson PPT and practice sheet.
Lesson 4 الحصة الرابعة **Genetic Engineering** الهندسة الوراثية	**Evaluate:** I can evaluate how technologies such as genetic engineering and gene therapy influence the inheritance of desired traits in organisms.	7.1 Increase engagement by offering choices and autonomy. 5.1 Provide multiple communication media (text, speech, visuals, models). 9.3 Develop self-assessment and self-reflection with tools, charts, and feedback displays.	Kindly check Toddle for the lesson PPT and practice sheet.
Homework الواجبات			Kindly check Toddle

Teacher's notes/ Reminders **ملاحظات المعلم والتذكير**	This week students learn how traits change over time: from mutations in genes (Lesson 1), to natural selection (Lesson 2), artificial selection (Lesson 3) and genetic engineering (Lesson 4). Please encourage your child to talk at home about one example from each lesson, such as an animal adapted to the UAE desert or a plant or animal bred by farmers.

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Technology Integration دمج التكنولوجيا	Videos, the PhET Natural Selection simulation, and Toddle for practice sheets and feedback.
Cross-curricular links الروابط بين المواد الدراسية	Geography (UAE ecosystems and desert habitats), Individuals and Societies (ethics of gene technology), and Biology.
External Assessment Integration دمج التقييمات الخارجية	Aligned to NGSS MS-LS3-1, MS-LS4-4 and MS-LS4-5 and MAP Growth Science (Life Sciences).
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Critical thinking (analysing evidence), collaboration (group work and peer feedback), communication (explaining ideas with scientific vocabulary).
Standards المعايير	• **MS-LS3-1:** Develop and use a model to describe why structural changes to genes (mutations) may affect proteins and result in harmful, beneficial or neutral effects. • **MS-LS4-4:** Construct an explanation of how genetic variations of traits increase some individuals' probability of surviving and reproducing. • **MS-LS4-5:** Gather and synthesize information about technologies that have changed the way humans influence the inheritance of desired traits.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How do changes in genes and traits influence the survival of living things over time?
Success Criteria معايير النجاح **(Measurable)**	The student has met the learning targets by: لقد حقق الطالب/الطالبة أهداف التعلم من خلال: **Lesson 1 – Genes, Mutations and Traits** • **MUST (DOK 1):** I can define gene, protein and mutation, and identify substitution, insertion and deletion mutations. • **SHOULD (DOK 2):** I can explain how a change in the DNA of a gene changes the protein and the trait. • **COULD (DOK 3):** I can justify, using evidence from an example, whether a mutation is harmful, beneficial or neutral. **Lesson 2 – Natural Selection** • **MUST (DOK 1):** I can define variation, adaptation and natural selection. • **SHOULD (DOK 2):** I can explain the steps of natural selection (variation, competition, survival, reproduction) using an example. • **COULD (DOK 3):** I can use data to predict how a population will change over several generations when its environment changes. **Lesson 3 – Artificial Selection** • **MUST (DOK 1):** I can define artificial selection (selective breeding) and identify two examples. • **SHOULD (DOK 2):** I can explain the similarities and differences between natural and artificial selection using a Venn diagram. • **COULD (DOK 3):** I can evaluate the benefits and drawbacks of selective breeding, such as the loss of genetic variation.

	Lesson 4 – Genetic Engineering • **MUST (DOK 1):** I can define genetic engineering and identify two examples, such as insulin-making bacteria and genetically modified crops. • **SHOULD (DOK 2):** I can explain how genetic engineering differs from artificial selection. • **COULD (DOK 3):** I can evaluate the benefits, risks and ethical questions of gene technologies, including genetic testing in the UAE.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Conservation الحفاظ علي البيئة Global Understanding التفاهم العالمي Heritage التراث
	Area: المجال Biodiversity, conservation and the adaptation of living things in the UAE, and the role of genetics in protecting the health of future generations.
	Implementation: التطبيق • Lesson 2: students explain how adaptations help UAE desert animals, such as the Arabian oryx, survive and reproduce. • Lesson 3: students link artificial selection to UAE heritage, such as the breeding of camels, Arabian horses and date palms. • Lesson 4: students discuss the UAE's premarital genetic testing and the Emirati Genome Programme as examples of gene technology protecting the community.